The Sender ID and Recipient ID are short sequences of bytes and are typically only allowed to be a few bytes long, so can't be globally unique names. Instead, they are used to distinguish the two parties involved in the communication. For example, some OSCORE implementations use a single zero byte for the client, and a 1 byte for the server. An optional ID Context string can be included in the Common Context, which can be used to map the Sender and Recipient IDs to device identities, for example in a lookup table.

The Master Key and Master Salt are combined using the HKDF key derivation function that you first used in chapter 10. Previously, you've only used the HKDF-Expand function, but this combination is done using the HKDF-Extract method that is intended for inputs that are not uniformly random. HKDF-Extract is shown in listing 13.7 and is just a single application of HMAC using the Master Salt as the key and the Master Key as the input. Open the HKDF.java file and add the `extract` method to the existing code.

Listing 13.7 HKDF-Extract

```
public static Key extract(byte[] salt, byte[]
inputKeyMaterial)    #A
        throws GeneralSecurityException {
    var hmac = Mac.getInstance("HmacSHA256");
    if (salt == null) {    #B
        salt = new byte[hmac.getMacLength()];    #B
    }    #B
    hmac.init(new SecretKeySpec(salt, "HmacSHA256"));    #C
    return new
SecretKeySpec(hmac.doFinal(inputKeyMaterial),    #C
            "HmacSHA256");
```